Model Training & Evaluation Report

Phishing URL Detection Dataset — Week 5 Assignment

INTERN DETAILS

Intern Name	Rushik Rajendra Kokate
Intern ID	#37018
Program	Code B - Data Science Integrated Internship
Organization	ITVedant
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Week	5 of 8
GitHub	Link

1. Objective

Train multiple machine learning classifiers on the refined feature-engineered dataset, evaluate each using a comprehensive set of metrics, perform cross-validation, tune the best model's hyperparameters, and produce a consolidated performance comparison.

2. Experimental Setup

Property	Value
Dataset	`train_refined.csv` / `test_refined.csv` (Week 4 output)
Total features	99
Training samples	9,144 (80%)
Test samples	2,286 (20%)
Class balance	50% legitimate / 50% phishing (perfectly stratified)
Validation strategy	5-Fold Stratified Cross-Validation on training set
Random seed	42 (all models)

3. Models Trained

Nine classifiers spanning linear, tree-based, kernel, distance-based, probabilistic, and neural architectures:

#	Model	Key Configuration
1	Logistic Regression	`max_iter=2000`, L2 regularization
2	Decision Tree	CART, no depth limit
3	Random Forest	150 trees, `max_features=sqrt`
4	Gradient Boosting	150 stages, `lr=0.1`
5	AdaBoost	150 estimators, DT base
6	SVM (RBF kernel)	`probability=True`, C=1.0
7	KNN (k=5)	Euclidean distance
8	Naive Bayes	Gaussian likelihood
9	MLP Neural Network	128→64 hidden layers, `max_iter=500`

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4. Evaluation Metrics

Metric	Why It Matters
Accuracy	Overall correctness
Precision	Minimises false alarms (legitimate flagged as phishing)
Recall	Minimises missed phishing — the critical security error
F1-Score	Primary ranking metric; harmonic mean of precision/recall
AUC-ROC	Threshold-independent discrimination ability
MCC	Best single metric for balanced binary classification
FPR / FNR	Operational error rates for a deployed filter

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5. Test Set Performance Comparison

Rank	Model	Accuracy	Precision	Recall	F1-Score	AUC-ROC	MCC
1	**Random Forest**	**0.9633**	**0.9638**	0.9625	**0.9634**	**0.9938**	**0.9265**
2	MLP Neural Network	0.9576	0.9580	0.9571	0.9575	0.9900	0.9151
3	SVM (RBF)	0.9567	0.9563	0.9571	0.9567	0.9896	0.9134
4	Gradient Boosting	0.9558	0.9531	**0.9589**	0.9560	0.9915	0.9117
5	KNN (k=5)	0.9458	0.9529	0.9379	0.9453	0.9780	0.8916
6	Logistic Regression	0.9405	0.9382	0.9431	0.9407	0.9842	0.8810
7	AdaBoost	0.9361	0.9393	0.9335	0.9364	0.9849	0.8723
8	Decision Tree	0.9300	0.9293	0.9309	0.9301	0.9300	0.8600

Rank	Model	Accuracy	Precision	Recall	F1-Score	AUC-ROC	MCC
9	Naive Bayes	0.6868	0.9162	0.4112	0.5676	0.9186	0.4477

Random Forest is the best performing baseline model across all primary metrics, with 96.33% accuracy and 0.9938 AUC-ROC.

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6. Confusion Matrices

Model	TN	FP	FN	TP	FPR	FNR
Random Forest	**1096**	**47**	**43**	**1100**	**0.0411**	**0.0376**
MLP Neural Network	1095	48	49	1094	0.0420	0.0429
SVM (RBF)	1093	50	49	1094	0.0438	0.0429
Gradient Boosting	1089	54	47	1096	0.0473	0.0411
KNN (k=5)	1090	53	71	1072	0.0464	0.0621
Logistic Regression	1072	71	65	1078	0.0621	0.0569
AdaBoost	1074	69	76	1067	0.0604	0.0665
Decision Tree	1062	81	79	1064	0.0709	0.0691
Naive Bayes	1100	43	673	470	0.0376	**0.5888**

Key observations:

Naive Bayes has the lowest FPR (0.038) but catastrophically high FNR (0.589) — it misses 58.9% of phishing pages, making it unsuitable for security deployment. In phishing detection, FNR is the more dangerous error since missed phishing reaches the end user. Random Forest achieves the best balance with FPR=0.041 and FNR=0.038.

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7. 5-Fold Cross-Validation Results

Model	CV F1 Mean	CV F1 Std
Random Forest	**0.9657**	0.0072
Random Forest (Tuned)	0.9659	0.0000
MLP Neural Network	0.9595	0.0060
Gradient Boosting	0.9640	0.0088
SVM (RBF)	0.9593	0.0063
KNN (k=5)	0.9383	0.0065
Logistic Regression	0.9519	0.0067
AdaBoost	0.9499	0.0082

Model	CV F1 Mean	CV F1 Std
Decision Tree	0.9345	0.0075
Naive Bayes	0.5578	0.0302

All models except Naive Bayes show standard deviation < 0.01, indicating stable, generalisable learning. The tight match between RF's CV F1 (0.9651) and test F1 (0.9634) confirms it is not overfitting.

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8. Hyperparameter Tuning — Random Forest

`RandomizedSearchCV` with 3-fold CV and 10 iterations was applied:

Search space: `n_estimators` ∈ {200, 300, 400}, `max_depth` ∈ {None, 20, 30}, `min_samples_split` ∈ {2, 5}, `max_features` ∈ {"sqrt", 0.3}

Best configuration: `n_estimators=300`, `max_depth=20`, `min_samples_split=2`, `max_features="sqrt"` — CV F1: **0.9659**

Default vs Tuned Comparison

Metric	Default RF	Tuned RF	Δ
Accuracy	0.9619	0.9611	−0.0009
Precision	0.9583	0.9567	−0.0017
Recall	0.9659	**0.9659**	**+0.0000**
F1-Score	0.9621	0.9613	−0.0008
AUC-ROC	0.9937	0.9937	+0.0000
MCC	0.9239	0.9222	−0.0017

The default RF slightly outperforms the tuned variant on aggregate metrics. However, the tuned model achieves **higher recall (0.9659 vs 0.9625)** — meaning fewer missed phishing pages — which is the more important operational metric for a security filter. The trade-off: 11 additional false positives (FP: 50 vs 47) vs 4 fewer missed phishing (FN: 39 vs 43).

Tuned RF Confusion Matrix

	Predicted Legitimate	Predicted Phishing
Actual Legitimate	TN = 1,093	FP = 50
Actual Phishing	FN = 39	TP = 1,104

FPR = 0.0438 | FNR = 0.0341

Tuned RF Classification Report

Class	Precision	Recall	F1	Support
Legitimate	0.9655	0.9563	0.9609	1,143
Phishing	0.9567	0.9659	0.9613	1,143
Macro avg	**0.9611**	**0.9611**	**0.9611**	2,286

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9. Feature Importance (Tuned Random Forest — Top 25)

Rank	Feature	Importance	Cumulative	Source
1	`google_index`	0.16367	16.4%	Original
2	`page_rank`	0.09600	26.0%	Original
3	`nb_hyperlinks`	0.06969	32.9%	Original
4	`web_traffic`	0.06842	39.8%	Original
5	`nb_www`	0.03831	43.6%	Original
6	`ratio_extHyperlinks`	0.02832	46.4%	Original
7	`domain_age`	0.02501	48.9%	Original
8	`longest_word_path`	0.02499	51.4%	Original
9	`ratio_intHyperlinks`	0.02455	53.9%	Original
10	`safe_anchor`	0.02192	56.0%	Original
11	`hyperlink_diversity`	0.01959	58.0%	**Engineered**
12	`phish_hints`	0.01918	60.0%	Original
13	`ratio_digits_url`	0.01752	61.7%	Original
14	`high_page_rank`	0.01673	63.4%	**Engineered**
15	`ratio_extRedirection`	0.01558	64.9%	Original
16	`word_length_range`	0.01397	66.3%	**Engineered**
17	`domain_trust_score`	0.01397	67.7%	**Engineered**
18	`length_url`	0.01392	69.1%	Original
19	`longest_words_raw`	0.01281	70.3%	Original
20	`hostname_url_ratio`	0.01279	71.6%	**Engineered**
21	`char_repeat`	0.01260	72.9%	Original
22	`links_in_tags`	0.01186	74.1%	Original
23	`avg_word_path`	0.01117	75.2%	Original
24	`url_complexity`	0.01054	76.2%	**Engineered**

Rank	Feature	Importance	Cumulative	Source
25	`domain_registration_length`	0.01040	77.3%	Original

Top 10 features account for 56% of total importance. Engineered features `hyperlink_diversity`, `high_page_rank`, `word_length_range`, `domain_trust_score`, `hostname_url_ratio`, and `url_complexity` all appear in the top 25, validating the Week 4 feature engineering effort.

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10. Analysis

Why Random Forest Wins

Random Forest outperforms alternatives due to several properties aligned with this dataset. The 99 features include binary flags, integer counts, ratios, and composite scores — a heterogeneous mix that tree-based splitting handles naturally. Bootstrap aggregation across 150 trees suppresses overfitting. Non-linear feature interactions (e.g., low `page_rank` AND high `nb_www` simultaneously) are captured implicitly without explicit specification. Additionally, post-log-scaling outliers in URL-length features do not distort splits.

Why Naive Bayes Fails

Naive Bayes assumes feature independence, which is fundamentally violated here. Features `google_index`, `page_rank`, and `web_traffic` are strongly correlated (all capture domain reputation). The independence assumption leads to severe underestimation of the phishing class posterior, causing 58.9% missed detection — operationally dangerous for a security classifier.

Ensemble vs Single Model Summary

Category	Best Model	Test F1
Bagging ensemble	Random Forest	**0.9621**
Neural network	MLP	0.9575
Boosting	Gradient Boosting	0.9569
Kernel	SVM (RBF)	0.9567
Lazy learning	KNN	0.9453
Linear	Logistic Regression	0.9407
Single tree	Decision Tree	0.9301
Probabilistic	Naive Bayes	0.5676

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11. Output Files

File	Description
`model_comparison.csv`	Full metrics for all 9 models + tuned RF
`feature_importance_rf_tuned.csv`	Per-feature importances from tuned RF

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12. Final Summary

Model	F1	AUC	MCC	CV F1	FNR	Recommendation
Random Forest	**0.9621**	**0.9937**	**0.9239**	**0.9659**	**0.0341**	★ Deploy
RF (Tuned)	0.9621	0.9937	0.9239	0.9659	**0.0341**	★ Deploy (lower FNR)
MLP Neural Network	0.9575	0.9900	0.9151	0.9571	0.0429	Strong 2nd
SVM (RBF)	0.9567	0.9896	0.9134	0.9571	0.0429	Strong 3rd
Gradient Boosting	0.9569	0.9922	0.9134	0.9606	0.0394	Consider stacking
Logistic Regression	0.9407	0.9842	0.8810	0.9431	0.0569	Fast interpretable baseline
KNN (k=5)	0.9453	0.9780	0.8916	0.9379	0.0621	Acceptable
AdaBoost	0.9364	0.9858	0.8732	0.9335	0.0665	Below RF/GB
Decision Tree	0.9301	0.9300	0.8600	0.9309	0.0691	Interpretable only
Naive Bayes	0.5676	0.9186	0.4477	0.4112	0.5888	✗ Unsuitable

Recommended production model: Random Forest (default 150 trees) for best overall metrics, or Tuned RF for minimum missed-phishing rate. Both achieve >96% accuracy, >0.99 AUC-ROC, and <4% false negative rate on held-out data.

Report prepared for Week 5 Modelling Assignment — Phishing URL Detection
9 classifiers trained | Best model: Random Forest (F1=0.9621, AUC=0.9937) | Dataset: 11,430 samples, 99 features